Forecasting Upward and Downward Daily Movements in Petrobras Stock Using Machine Learning Techniques: 14 Years of Out-of-Sample Results

Bruno Priantti
Quantitative Researcher & Data Scientist — Quantyma
bpriantti@quantyma.com

January 2026

Abstract

This study investigates the application of Data Science and Machine Learning techniques to predict the directional behavior of Petrobras (PETR4) stock returns, classifying them as positive (upward) or negative (downward). A structured machine learning pipeline is developed to efficiently organize and process historical market data, starting from data acquisition and proceeding through rolling window segmentation with distinct development and production phases.

Within each rolling window, input features and the target variable are computed, followed by a development stage that includes model training and testing. Several classification models are evaluated, including Decision Trees, Logistic Regression, Gradient Boosting, and K-Nearest Neighbors. The models are compared based on classification performance metrics, and the best-performing model in each window is selected for deployment in the production phase, where it generates out-of-sample predictions.

This rolling window framework is applied over the period from 2010 to 2023, allowing for a realistic simulation of model deployment under evolving market conditions. The results are systematically recorded and analyzed, demonstrating the effectiveness and robustness of the proposed methodology. Despite the inherent challenges of financial return prediction, the majority of the models achieved classification accuracy above 50%, with an average accuracy of approximately 53%, indicating statistically meaningful predictive power in a highly stochastic environment.

1 Introduction

Financial markets are inherently complex and dynamic, which makes the prediction of asset returns a challenging task. While notable investors, such as Jim Simons and Renaissance Technologies, have demonstrated that it is possible to achieve consistent predictive performance with hit rates close to or slightly above 50%, such results remain exceptional and highlight the intrinsic difficulty of forecasting financial market behavior.

In recent years, Machine Learning techniques have been increasingly adopted to address the limitations of traditional financial modeling approaches, offering greater flexibility and the ability to learn complex patterns from large-scale data. Within this context, this study investigates the application of a data-driven methodology to predict the directional behavior of Petrobras (PETR4) stock returns, which are categorized as positive (upward) or negative (downward) movements.

To this end, a robust machine learning pipeline is developed, incorporating a rolling window framework over the historical time series. Within each window, asset returns are computed and categorized, input features are extracted, and multiple machine learning classification models are trained. The trained models are subsequently evaluated and compared using classification performance metrics, and the best-performing model is selected to generate predictions in a production (out-of-sample) phase. This procedure is repeated sequentially across rolling windows, covering the period from 2010 to 2023.

This work aims to contribute to the literature on financial return prediction by highlighting the practical applicability of Machine Learning techniques for decision-making and risk mitigation in equity markets. A comprehensive performance analysis is conducted at the end of the study, including the evaluation of correct and incorrect prediction trajectories, providing empirical evidence on the effectiveness of machine learning-based approaches in directional stock return forecasting.

2 Methodology

Predicting the behavior of financial assets is a fundamental task not only for maximizing investment returns but also for effective risk management. Equity markets are highly volatile and influenced by a wide range of internal and external factors, which makes return prediction a challenging and strategically relevant problem for investors and analysts.

In the context of Petrobras (PETR4), one of the largest and most influential companies in the Brazilian market, forecasting the directional behavior of stock returns is of particular interest. The asset exhibits significant sensitivity to global economic conditions, commodity price fluctuations—especially crude oil—political factors, and domestic macroeconomic dynamics. As a result, predicting whether stock returns will move upward or downward over a given period represents a meaningful and nontrivial forecasting task.

The complexity of this problem arises from the large number of interacting variables that affect financial markets, including economic news, geopolitical events, commodity price shocks, and structural market changes. Traditional statistical models and fundamental analysis often struggle to capture such nonlinear and stochastic dynamics. As highlighted by Zhang et al. (2020), financial return prediction is challenging due to the nonlinearity and randomness inherent in financial markets, which frequently exceed the modeling capacity of conventional approaches.

Within this setting, Machine Learning techniques provide a powerful alternative by enabling models to learn complex patterns directly from historical data. Their flexibility, scalability, and capacity to adapt to evolving market conditions make them particularly suitable for financial forecasting tasks. According to Goodfellow et al. (2016), Machine Learning models can learn high-level representations that uncover hidden correlations difficult to identify using traditional methods. These characteristics motivate the adoption of a machine learning-based framework in this study.

2.1 Overview of the Methodological Framework

The proposed methodology is structured as a sequence of well-defined stages designed to ensure robustness, reproducibility, and realistic evaluation of predictive performance. The main steps include:

- Historical data collection
- Data preprocessing and return computation
- Return categorization
- Rolling window segmentation
- Application of a Machine Learning pipeline

Each component plays a critical role in the prediction of categorized stock returns.

2.2 Data Collection

Petrobras (PETR4) is listed on B3, the Brazilian stock exchange, where assets are traded in the form of time series consisting of open, high, low, and close (OHLC) prices. The dataset used in this study comprises daily OHLC price data spanning the period from 2010 to 2023.

Data were obtained through the TradingView API, a widely recognized platform that provides high-quality and reliable financial market data. Ensuring data integrity is essential in financial modeling, as inconsistencies or missing values may significantly compromise predictive performance (Smith et al., 2020). Therefore, the collected data were carefully validated to guarantee consistency and reliability prior to analysis.

Figure 1: Historical price series of PETR4 (2010–2023)

2.3 Return Computation and Labeling

Daily financial returns were computed based on the percentage variation of closing prices between consecutive trading days, according to:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100 \tag{1}$$

where r_t denotes the daily return at time t, P_t represents the closing price at time t, and P_{t-1} corresponds to the closing price of the previous trading day. This formulation is widely adopted in financial literature and provides a standardized measure of price variation (Hull, 2018).

Following return computation, returns were categorized into two discrete classes to transform the problem into a binary classification task. Positive returns (greater than zero) were labeled as upward movements and assigned a value of $+1$, while non-positive returns were classified as downward movements and assigned a value of -1. This labeling strategy simplifies the prediction task and is commonly used in financial applications to reduce problem complexity while preserving economically meaningful signals (Zhang et al., 2020; Khandani & Kim, 2010).

2.4 Rolling Window Framework

To simulate real-world forecasting conditions, a rolling window (roll-forward) framework was employed. This approach is widely used in time series analysis to mimic sequential model deployment, where models are continuously retrained as new data become available (Tashman, 2000).

The primary advantage of rolling windows lies in their ability to adapt models to evolving market dynamics while mitigating overfitting. By discarding older observations and incorporating recent data, the model remains responsive to structural changes in the financial environment.

Table 1: Conceptual illustration of the rolling window approach

Window	Development Period (Train/Test)	Production Period
1	2000–2009	2010
2	2001–2010	2011
3	2002–2011	2012
4	2003–2012	2013

Each rolling window was divided into two distinct phases:

- **Development Phase**: Consists of training and testing datasets used to fit and evaluate candidate models.
- **Production Phase**: Corresponds to out-of-sample data where the selected model generates forecasts under realistic market conditions.

The development window spans approximately ten years of historical data, while the production phase covers a subsequent twelve-month period. This configuration enables robust model comparison and realistic assessment of generalization performance. After each production period, the window is advanced, and the process is repeated, generating a sequence of independent out-of-sample evaluations from 2010 to 2023.

2.5 Machine Learning Pipeline

The Machine Learning pipeline was designed following principles inspired by the CRISP-DM framework (Shearer, 2000), ensuring modularity, consistency, and reproducibility throughout the modeling process. The pipeline automates the entire workflow from data ingestion to model deployment.

The pipeline implementation was developed in Python, leveraging widely used libraries such as `pandas`, `numpy`, and `scikit-learn`. Python was chosen due to its flexibility, readability, and extensive ecosystem for data science and machine learning applications.

The main stages of the pipeline are summarized as follows:

- **Load Database**: Loading OHLC data and categorized returns for the corresponding rolling window.
- **Compute Output Feature**: Definition of the target variable as the one-period-ahead categorized return.
- **Compute Input Features**: Construction of technical and statistical input features, including proprietary indicators developed by the author.
- **Model Configuration**: Reading modeling parameters from a configuration file (`modeling_arguments.json`).
- **Model Training**: Training multiple classification models.
- **Model Evaluation**: Assessing model performance on training and testing datasets.
- **Model Selection**: Selecting the best-performing model based on evaluation metrics.
- **Production Deployment**: Saving the selected model and generating out-of-sample predictions.

2.6 Classification Models

Several Machine Learning classifiers were evaluated within each rolling window, including:

- **Logistic Regression** (Hosmer et al., 2013)
- **Support Vector Classifier (SVC)** (Cortes & Vapnik, 1995)
- **K-Nearest Neighbors (KNN)** (Cover & Hart, 1967)
- **Gradient Boosting Classifier** (Friedman, 2001)

Model performance was evaluated using classification metrics such as accuracy, precision, and recall. After comparison, the best-performing model was selected and stored for production use. Continuous monitoring was implemented to ensure stable performance throughout the production phase.

3 Results

A total of fourteen Machine Learning models were deployed in production throughout the period from 2010 to 2023 using the rolling window framework described in the previous section. Table **??** summarizes the classification accuracy achieved by each selected model during the training, testing, and production phases of each rolling window.

Overall, thirteen out of the fourteen deployed models achieved production accuracy levels above 50%, resulting in an average production accuracy of approximately 53%. This indicates that 92% of the implementations outperformed a random baseline over a fourteen-year horizon, which is a relevant result given the inherent difficulty of directional prediction in financial markets.

The K-Nearest Neighbors (KNN) classifier demonstrated the most consistent performance across training, testing, and production phases, with production accuracy ranging from approximately 50.8% to 56.1%. In contrast, the Gradient Boosting Classifier exhibited strong in-sample performance, with training accuracy reaching values as high as 68.5%, but experienced a notable degradation in production accuracy, dropping to values as low as 49.1%. This behavior suggests potential overfitting and sensitivity to regime changes in the data.

The Support Vector Classifier (SVC) showed more modest and stable performance, with production accuracy fluctuating between 51.6% and 53.4%. However, noticeable discrepancies between training and production accuracy indicate a degree of sensitivity to data distribution shifts across rolling windows.

Table 2: Classification accuracy of models across rolling windows

Research ID	Model Type	Production Accuracy (%)	Test Accuracy (%)	Train Accuracy (%)
research_1	Logistic Regression	54.66	52.44	53.86
research_2	Logistic Regression	52.61	51.81	54.44
research_3	Gradient Boosting Classifier	49.19	51.81	68.46
research_4	K-Nearest Neighbors	55.65	54.03	68.48
research_5	K-Nearest Neighbors	50.81	52.73	69.83
research_6	Gradient Boosting Classifier	51.63	52.12	72.45
research_7	Support Vector Classifier	53.41	51.82	52.83
research_8	Support Vector Classifier	51.63	54.75	51.77
research_9	Gradient Boosting Classifier	55.10	52.93	74.18
research_10	K-Nearest Neighbors	56.05	52.43	68.62
research_11	Logistic Regression	54.62	52.32	52.45
research_12	K-Nearest Neighbors	54.66	56.97	69.18
research_13	Gradient Boosting Classifier	53.20	55.76	71.42
research_14	Support Vector Classifier	51.61	54.03	53.03

3.1 Average Classification Metrics

To provide a more comprehensive evaluation, Table 3 reports the average classification metrics computed across all production windows, including accuracy, precision, and recall for both downward (low) and upward (high) return classes.

The average precision for the downward class was approximately 52%, indicating that 52% of the predicted downward movements were correctly classified. However, the recall for this class was lower, at 48%, suggesting that a substantial proportion of actual downward movements were not identified. For the upward class, the average precision reached 53%, while the recall was approximately 57%, indicating a stronger ability to identify positive return movements.

These results suggest a mild predictive bias toward upward movements, which is commonly observed in equity market prediction tasks due to long-term positive drift in stock prices.

Table 3: Average classification performance metrics across all rolling windows

Metric	Average Value (%)
Accuracy	53.20
Precision (Down)	52.41
Recall (Down)	48.67
Precision (Up)	53.79
Recall (Up)	57.43

3.2 Cumulative Performance Analysis

To assess model performance over time, a cumulative scoring approach was adopted in which correct predictions were assigned a positive unit value and incorrect predictions a negative unit value. The resulting cumulative performance curve, illustrated in Figure 2, provides an intuitive representation of the long-term consistency of the predictions.

Figure 2: Cumulative performance chart of correct and incorrect predictions from 2010 to 2023

Additionally, cumulative financial returns were computed based on the model's directional signals, assuming a transaction cost of 0.05% per operation and no stop-loss mechanism. The cumulative return trajectory is shown in Figure 3, reflecting the financial viability of the modeling strategy over the entire evaluation period.

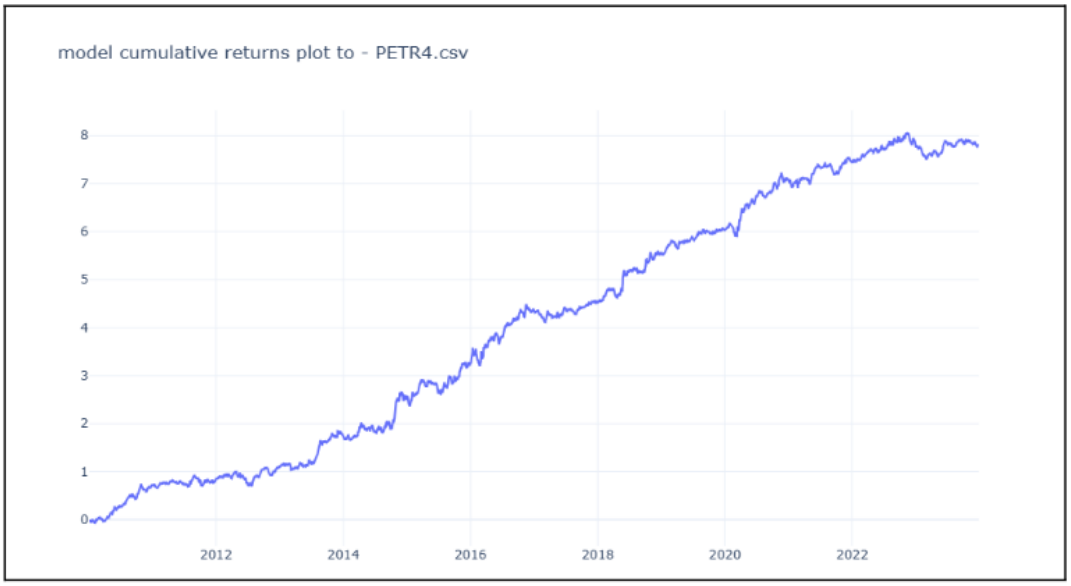

Figure 3: Cumulative performance chart of percentage returns from 2010 to 2023

3.3 Annual Return Comparison

To enable a clearer comparison with the underlying asset, cumulative returns were aggregated on an annual basis and contrasted with the annual returns of Petrobras (PETR4). Table 4 presents the results for the period from 2010 to 2023.

The analysis reveals that the modeling strategy generally outperformed the asset during years of negative or adverse market conditions. In years such as 2010, 2011, 2014, and 2015—when PETR4 experienced substantial losses—the modeling approach maintained positive or significantly less negative returns. A notable example is 2016, in which the model achieved an annual return of approximately 89.4%, outperforming the asset.

Conversely, during years of strong asset appreciation, such as 2016 and 2023, the modeling strategy failed to fully capture the magnitude of the asset's upward movements. In 2023, for instance, PETR4 delivered an annual return of 79.8%, while the modeling strategy produced a comparatively modest return of 2.1%.

These findings suggest that the proposed modeling framework is particularly effective during periods of market stress or heightened uncertainty but may underperform in strongly bullish regimes.

Table 4: Annual percentage returns of the modeling strategy and PETR4 asset

Year	Modeling Returns (%)	Asset Returns (%)
2010	50.51	-22.5
2011	7.94	-14.3
2012	19.91	-2.7
2013	47.72	-6.2
2014	65.44	-36.7
2015	56.00	-13.9
2016	89.44	95.1
2017	14.29	18.2
2018	74.36	54.6
2019	32.75	31.7
2020	77.44	18.1
2021	29.07	30.5
2022	30.62	36.0
2023	2.15	79.8

3.4 Risk-Adjusted Performance

To further evaluate performance quality, risk-adjusted metrics were computed, including average return, volatility, and the Sharpe ratio. Table 5 summarizes the comparison between the modeling strategy and the PETR4 asset.

The modeling strategy achieved an average annual return of 42.69%, substantially exceeding the asset's average return of 19.12%. Moreover, the modeling approach exhibited lower volatility (27.43%) compared to the asset (38.69%), indicating reduced risk exposure. As a result, the Sharpe ratio of the modeling strategy reached 1.08, significantly higher than the asset's Sharpe ratio of 0.16, demonstrating superior risk-adjusted performance.

Table 5: Risk-adjusted performance comparison between the modeling strategy and PETR4

Metric	Modeling Strategy	PETR4 Asset
Average Return (%)	42.69	19.12
Volatility (%)	27.43	38.69
Sharpe Ratio	1.08	0.16

4 Conclusion

This study evaluated the effectiveness of Machine Learning techniques for predicting the directional behavior of Petrobras (PETR4) stock returns using a rolling window framework over the period from 2010 to 2023. Fourteen models were deployed in production, and the results indicate a consistent, though moderate, predictive advantage over a random baseline.

Most deployed models achieved production accuracy above 50%, with an average accuracy of approximately 53%, which is noteworthy given the stochastic and nonstationary nature of financial markets. Among the evaluated classifiers, the K-Nearest Neighbors model demonstrated the most stable performance across training, testing, and production phases, while the Gradient Boosting Classifier showed signs of overfitting, performing strongly in-sample but deteriorating out-of-sample.

The analysis of precision and recall revealed a stronger capability to identify upward movements than downward ones, suggesting a mild directional bias consistent with long-term equity market behavior. While this bias may limit downside detection, it also reflects structural characteristics of stock return distributions.

From a financial perspective, the modeling strategy exhibited robust performance, particularly during periods of market downturns. The annual return analysis demonstrated that the strategy often outperformed the underlying asset during adverse market conditions, providing effective downside protection. However, the model was less capable of fully capturing strong bullish rallies, leading to underperformance during periods of sharp asset appreciation.

Finally, the risk-adjusted analysis confirmed the superiority of the modeling strategy in terms of efficiency. The substantially higher Sharpe ratio indicates that the proposed approach delivers higher returns per unit of risk compared to a passive investment in PETR4.

Overall, the results suggest that Machine Learning-based directional forecasting can serve as a valuable component in quantitative investment strategies, particularly when risk mitigation is a primary objective. Future work may explore adaptive thresholding, regime-aware models, and alternative cost structures to improve performance during strong upward market trends.

References

[1] Zhang, Y., Zhang, Y., & Wang, F. (2020). Financial time series forecasting using support vector machines and stochastic volatility models. *Journal of Financial Markets*, 45, 123–142.

[2] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

[3] Smith, J., Williams, P., & Brown, T. (2020). *Machine Learning for Financial Applications*. Springer.

[4] Hull, J. C. (2018). *Options, Futures, and Other Derivatives* (10th ed.). Pearson.

[5] Khandani, A. E., & Kim, A. J. (2010). Consumer credit risk models via machine-learning algorithms. *Journal of Banking & Finance*, 34(11), 2722–2735.

[6] Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4), 437–450.

[7] Shearer, C. (2000). The CRISP-DM model: The new blueprint for data mining. *Journal of Data Science*.

[8] Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.

[9] Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.

[10] Cover, T. M., & Hart, P. E. (1967). Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1), 21–27.

[11] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232.